

Ph.D. Scholar in Computer Science
Marie Curie Fellow

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Collaborative Networking Lab
Department of Computer Science
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Education

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| 2015 – | ■ Ph.D. Scholar
– Supervised by <i>Prof. Dr. Jussi Kangasharju</i>
– Collaborative Networking (CoNe) research group | University of Helsinki
Finland |
| 2013 – 2015 | ■ M.Tech. Computer Science & Engineering
– Specialization in <i>Mobile Computing</i>
– CGPA: 8.5/10
– Nominated for Best Thesis Award | IIIT Delhi
India |
| 2009 – 2013 | ■ B.Tech. Computer Engineering
– First Division <i>with Distinction</i>
– Ranked amongst Top-10 batch performers | Punjabi University
India |

Skills

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| Research Interests | ■ Edge clouds, Cloud computing, Internet-of-Things, Multipath TCP, Information-Center Networks, Distributed Systems |
| Programming | ■ Python, Java, HTML5, C, C++, BASH, L ^A T _E X |
| Tools & Technologies | ■ Linux kernel programming, Network Simulator (NS-3), MATLAB, SPARK, Docker, GRAPHLAB, TENSORFLOW |
| Languages | ■ English Expert Hindi Expert Punjabi Expert Finnish Beginner |

Research Experience

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| Ph.D. | ■ University of Helsinki
Supervisor: <i>Prof. Dr. Jussi Kangasharju</i>
– Research in operation, deployment & networking in edge clouds
– Jointly funded by EU FP7 project – CleanSky ITN & Finnish Academy of Sciences project – AIDA | Helsinki, Finland
Sept 2015 – June 2019 (expected) |
| Visiting Researcher | ■ University of Göttingen
Host: <i>Prof. Dr. Xiaoming Fu</i>
– Research in optimal placement of edge servers and virtualized service functions
– Funded by EU FP7 project – CleanSky ITN | Göttingen, Germany
Jan – May 2018 |

Research Experience (continued)

	■ NEC research labs, Europe Host: <i>Dr. Fabian Schneider</i> – Research in deploying network functions in edge clouds – Funded by EU FP7 project – CleanSky ITN	Heidelberg, Germany Nov 2016 – Mar 2017
Project Scientist	■ Indian Inst. of Technology (IIT) Supervisor: <i>Dr. Vinay Ribeiro</i> – Research & Development in "Structured sharing of resources in a Device-to-Device (D2D) Network" – Funded by Intel Labs, USA	New Delhi, India June – Aug 2015
M.Tech. (with Thesis)	■ Indraprastha Inst. of Information Tech. (IIIT) Supervisor: <i>Dr. Pushpendra Singh</i> – Research in checkpointing user-applications in Information-centric networks – Awarded GATE scholarship by Govt. of India	New Delhi, India 2013 – 2015
Research Intern	■ Central Scientific Instruments Org. (CSIO) Supervisor: <i>Dr. Sudesh K. Mittal</i> – Research in landslide prediction model using image processing and interpolation – Collaboration with Indian Space Research Org. (ISRO)	Chandigarh, India Jan – July 2013

Teaching Experience

Guest Instructor	■ Cloud & Edge Computing (University of Helsinki)	2019
Teaching Assistant	■ Distributed Data & Infrastructures (University of Helsinki)	2017, 2018
	■ Distributed Systems (University of Helsinki)	2016
	■ Cellular Data Networks (IIIT Delhi)	2015

Positions of Responsibility

Shadow PC	■ ACM Internet Measurement Conference (IMC)	2019
External Reviewer	■ IEEE ICC NGNI Symposium	2019
	■ ACM Information Centeric Networking Conference (ICN)	2017
Jury	■ HackEdge hackathon organized by CEACS	2017
Organizer	■ 3 rd International CleanSky Conference	2017
	■ 2 nd International CleanSky Summer School	2016
Web-Admin	■ 2 nd International CleanSky Summer School	2016
Head	■ Design Team of IIITD Research Showcase	2015
Representative	■ M.Tech. 2013-15 batch at IIIT Delhi	2013, 2014
	■ Post-Graduate Disciplinary Action Committee at IIIT Delhi	2013, 2014

Awards & Achievements

- 1 Awarded [SIGCOMM travel grant](#) to attend IMC 2019 Shadow PC meeting held in London, UK
- 2 Awarded [Microsoft Research travel grant](#) for ACM MOBICOM 2018 held in New Delhi, India
- 3 Awarded EU FP7 [Marie Curie Initial Training Network \(ITN\)](#) funding to support Ph.D. studies
- 4 Invited to attend [South Asia Workshop On Research Frontiers In Computing](#) with full-fellowship held at National University of Singapore (NUS), Singapore in May, 2014
- 5 [All-India percentile 98.57](#) in Graduate Aptitude Test Examination (GATE) – 2014 amongst ~2 million candidates
- 6 Awarded [GATE scholarship](#) by Govt. of India from 2013 – 2015
- 7 [Recognition Award](#) by Director, IIIT Delhi for exceptional service to Institute from 2013 – 2015
- 8 Graduated Bachelors with [1st Division with Distinction](#) and ranked amongst top 10 students
- 9 Awarded [Certificate of Merit](#) by Govt. of India for being [All-India Topper](#) in Mathematics in All-India Secondary School Examination in 2007

Research Projects

AIDA	 Adaptive and Intelligent Data – Objective : Introducing learning and intelligence in networks – PI : <i>Prof. Dr. Jussi Kangasharju</i> (Univ. of Helsinki) <i>Prof. Dr. Aristides Gionis</i> (Aalto Univ.)	Academy of Finland 2018–2021
BCDC	 Bright Clouds Dark Clouds www.bcdcenergia.fi – Objective : Digitalizing utilization of renewable energy via Cloud – PI : <i>Prof. Dr. Rauli Svento</i> (Univ. of Oulu)	Academy of Finland 2016–2020
CleanSky	 CleanSky Initial Training Network www.cleansky-itn.eu – Marie Curie Framework Program (FP) 7 Project – Objective : Innovation in several areas of cloud and edge computing – Collaboration : Nine university and industry partners in Europe, USA & China – PI : <i>Prof. Dr. Xiaoming Fu</i> (Univ. of Göttingen)	European Union 2014–2018

Research Contribution

Journal Publications

2018	 Placing it right!: Optimizing Energy, Processing, & Transport in Edge-Fog clouds N. Mohan & J. Kangasharju International Journal on <i>Annals of Telecommunication</i> , Springer, 2018 Paper: https://doi.org/10.1007/s12243-018-0649-0	Springer
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Research Contribution (continued)

Conference & Workshop Publications

- 2019 ■ [In Review] Pruning Edge Research with Latency Shears HotNets '19
N. Mohan, A. Zavodovski, W. Wong, P. Zhou & J. Kangasharju
18th ACM Workshop on *Hot Topics in Networks*
- [In Review] Cumulus: Unified Cloud Computing as a Commodity CoNEXT '19
W. Wong, A. Zavodovski, P. Zhou, N. Mohan, & J. Kangasharju
15th ACM Conference on emerging Networking EXperiments and Technologies
- [In Review] Is two greater than one?: Analyzing Multipath TCP over dual-LTE in the Wild IMC '19
N. Mohan, T. Shreedhar, A. Zavodovski, J. Kangasharju & S.K. Kaul
ACM Internet Measurement Conference
- Open Infrastructure for Edge: A Distributed Ledger Outlook HotEdge '19
A. Zavodovski, N. Mohan, W. Wong & J. Kangasharju
USENIX ATC workshop on *Hot Topics in Edge Computing*
- DeCloud: Truthful Decentralized Double Auction for Edge Clouds ICDCS '19
A. Zavodovski, S. Bayhan, N. Mohan, P. Zhou, W. Wong & J. Kangasharju
39th International Conference on Distributed Computing Systems
- ExEC: Elastic Extensible Edge Cloud EdgeSys '19
A. Zavodovski, N. Mohan, S. Bayhan, W. Wong & J. Kangasharju
ACM EuroSys workshop on *Edge Systems, Analytics & Networking*
Paper: <https://doi.org/10.1145/3301418.3313941>
Slides: <http://bit.ly/2YGZ3jD>
- 2018 ■ ICON: Intelligent Container Overlays HotNets '18
A. Zavodovski, N. Mohan, S. Bayhan, W. Wong & J. Kangasharju
17th ACM Workshop on *Hot Topics in Networks*
Paper: <https://doi.org/10.1145/3286062.3286065>
Slides: <https://bit.ly/2UMcAUE> Video: <https://youtu.be/BNmF0gTV1R4>
- Anveshak: Placing Edge Servers In The Wild MECOMM '18
N. Mohan, A. Zavodovski, P. Zhou & J. Kangasharju
ACM SIGCOMM workshop on *Mobile Edge Communications*
Paper: <https://doi.org/10.1145/3229556.3229560>
Slides: <https://bit.ly/2W9JWx3> Video: <https://youtu.be/8wYjQr80iwI>
- QAware: A Cross-Layer Approach to MPTCP Scheduling IFIP Networking '18
T. Shreedhar, N. Mohan, S.K. Kaul & J. Kangasharju
IFIP Networking Conference, Zürich. May 2018
Paper: <https://doi.org/10.23919/IFIPNetworking.2018.8696843>
Open Source: <https://bit.ly/2Jod1Du>

Research Contribution (continued)

- MUTE: Multi-Tier Edge networks CrossCloud '18
*A. Silvestro**, *N. Mohan**, J. Kangasharju, F. Schneider & X. Fu [**joint authors*]
ACM EuroSys workshop on *CrossCloud Infrastructures & Platforms*
Paper: <https://doi.org/10.1145/3195870.3195871>
Slides: <https://bit.ly/2CxCjct>
- 2017 ■ Managing Data in Computational Edge Clouds MECOMM '17
N. Mohan, P. Zhou, K. Govindaraj & J. Kangasharju
ACM SIGCOMM workshop on *Mobile Edge Communications*
Paper: <https://doi.org/10.1145/3098208.3098212>
Slides: <https://bit.ly/2HJB5yn> Video: <https://youtu.be/akJtWokhrj4>
- 2016 ■ Edge-Fog Cloud: A Distributed Cloud for Internet of Things Computations CIoT '16
N. Mohan & J. Kangasharju
IEEE conference on *Cloudification of the Internet of Things*
Paper: <https://doi.org/10.1109/CIOT.2016.7872914>
Slides: <https://bit.ly/2Hwdrpz>
- Posters & Demos**
- 2018 ■ Redesigning MPTCP for Edge Clouds MobiCom '18
N. Mohan, T. Shreedhar, A. Zavodovski, O. Waltari, J. Kangasharju & S.K. Kaul
ACM conference on *Mobile Computing and Networking*
Paper: <https://doi.org/10.1145/3241539.3267738>
Poster: <https://bit.ly/2TNfUCB> SRC slides: <https://bit.ly/2Fg1iSk>
- 2017 ■ Grouping Computational Data in Resource Caches of Edge-Fog Cloud CrossCloud '17
N. Mohan, P. Zhou, K. Govindaraj, & J. Kangasharju
ACM EuroSys workshop on *CrossCloud Infrastructures & Platforms*
Paper: <https://doi.org/10.1145/3069383.3069391>
Poster: <https://bit.ly/20h2qsJ>
- 2014 ■ CCNCheck: Enabling Efficient Checkpointed Applications in Content Centric Networks CCNxCon '14
N. Mohan & P. Singh
PARC International *Content Centric Networking conference*
Paper: <https://bit.ly/2CtWqIy>
- Thesis**
- 2019 ■ [In Review] Edge Computing Platforms & Protocols Ph.D.
Nitinder Mohan
Submitted at: *University of Helsinki, Finland. June 2019*

Research Contribution (continued)

- 2015 ■ Enabling Sender-initiated Distributed Applications and Checkpointing in Content Centric Networks M.Tech.
Best Thesis Nominee
Nitinder Mohan
Submitted at: *IIIT Delhi, India. May 2015*
Examiner: *Dr. Vinay Ribeiro (IIT Delhi) & Dr. Vikram Goyal (IIIT Delhi)*
Paper: <https://bit.ly/2TWyKAd>

Talks

- 2018 ■ Providing Networking at Edge CleanSky Workshop '18
4th International CleanSky Summer School
Host: Prof. Dr. Jussi Kangasharju
University of Helsinki, Finland
- 2017 ■ Scaling computational clouds to network edge CleanSky Summer School '17
3rd International CleanSky Summer School
Host: Prof. Dr. Dan Li
Tsinghua University, Beijing, China
- Filling Edge-Fog cloud cover: All power to robots! CleanSky Workshop '17
3rd International CleanSky Workshop
Host: Prof. Dr. Xiaoming Fu
University of Göttingen, Germany
- 2016 ■ *MiddleBox Broker*: Enabling in-network functions in secured traffic CleanSky Workshop '16
2nd International CleanSky Workshop
Host: Prof. Dr. Yuming Jiang
NTNU Trondheim, Norway
- *Edge-Fog cloud*: Scaling IoT computations on the Edge GI-Dagstuhl '16
Aware Machine-to-Machine Communication
GI-Dagstuhl Seminar 16353
Dagstuhl, Germany
- Computing IoT data on *Edge-Fog cloud* CleanSky Summer School '16
2nd International CleanSky Summer School
Host: Prof. Dr. Jussi Kangasharju
University of Helsinki, Finland
- Simplifying IoT computation in the “Cloud” CleanSky Workshop '16
1st International CleanSky Workshop
Host: Dr. Fabian Schneider
NEC Europe Labs, Heidelberg, Germany
- 2015 ■ Scaling eScience computational-intensive cloud CleanSky Summer School '15
1st International CleanSky ITN Summer School
Host: Prof. Dr. Xiaoming Fu
University of Göttingen, Germany

Conference Participation

International Conferences

- 2019 ■ Network Traffic Measurements & Analysis (TMA) Paris, France
- 2018 ■ ACM MOBICOM New Delhi, India
- ACM SIGCOMM Budapest, Hungary

Conference Participation (continued)

	■ IFIP Networking	Zürich, Switzerland
	■ EuroSys	Porto, Portugal
2017	■ Networked Systems	Göttingen, Germany
	■ EuroSys	Belgrade, Serbia
2016	■ ACM SIGCOMM	Florianopolis, Brazil
	■ IEEE Cloudification of IoT	Paris, France

Seminars/Summer Schools

2017	■ 3 rd CleanSky ITN Summer School nasp.cs.tsinghua.edu.cn/CleanSky/	Beijing, China
2016	■ [GI-Dagstuhl] Aware-to-aware machine communication www.dagstuhl.de/en/program/calendar/evhp/?seminr=16353	Dagstuhl, Germany
	■ 2 nd CleanSky ITN Summer School cleanskysummer.cs.helsinki.fi/	Helsinki, Finland
	■ Mobile Services and Edge Computing Workshop www.cs.helsinki.fi/group/close/edge-computing-2016/	Organizer Web-Chair Helsinki, Finland
2015	■ 1 st CleanSky ITN Summer School cleanskysummer.uni-goettingen.de/	Helsinki, Finland
2014	■ South Asia Workshop On Research Frontiers In Computing www.comp.nus.edu.sg/programmes/pg/workshops/southasia/	Singapore

Open Source Contributions

- [rpi-mptcp](#), A tool for measuring MPTCP performance over LTE links on Raspberry Pi
- [anveshak](#), A tool which calculates optimal deployment of cellular Edge servers in an area
- [ns3-qaware](#), ns3 implementation of MPTCP QAware scheduler
- [qaware](#), Linux kernel implementation of MPTCP QAware scheduler
- [efsim](#) A Python-based simulator for Edge cloud networks
- [ccnac](#), A tool to enable application checkpointing in Information-centric networks

Memberships

- Association of Computing Machinery (ACM) Student Member
- Institute of Electrical and Electronics Engineers (IEEE) Student Member
- SIGCOMM Membership
- EUROSYS Membership
- Indian Society of Technical Education (ISTE) Student Member

References

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